Go Beyond the Numbers: Translate Data into Insights

PACE: Plan Stage

- What are the data columns and variables and which ones are most relevant to your deliverable?

The data columns are video-duration-sec, video-transcription-text, author-ban-status, video-view-count, video-like-count, video-share-count, video-download-count, video-comment-count. The columns that are most relevant to my deliverables are the engagement metrics.   s the view count, like count, share count, download count, and comment count, the units are in counts

- What units are your variables in?

The video duration unit is seconds and the engagement metrics units are counts, which represents the number of the time each action was performed.

- What are your initial presumptions about the data that can inform your EDA, knowing you will need to confirm or deny with your future findings?

I presume that videos posted by verified accounts are most likely to be opinion, whereas videos posted by unverified accounts are most likely to be claims. I also suspect that there will be a strong correlation between claim_status and engagement metrics. During my EDA, I will inspect whether such relationships hold true.

- Is there any missing or incomplete data?

Yes, there are 298 missing values in `claim_status`, `video_transcription_text`,`video_view_count`, `video_like_count`, `video_share_count`, `video_download_count`, and `video_comment_count`.

- Are all pieces of this dataset in the same format?

No, there are categorical variables, such as the target variable “claim_status” and continuous variables, such as the engagement metrics.

- Which EDA practices will be required to begin this project?

The six steps of EDA are required to begin the project: Discovering: Examining the dimensionality of the datase, such as the shape and the size of the data. Structure: Transforming the raw data into usable layout. Cleaning: Addressing any missing values, checking for duplicatatoins, and handling outliers. Join: Merging different datasets, if needed, to access all relevant data needed for unbiased and accurate predictions. Validating: Verifying whether the cleaning process leads to consistent and high quality data. Presenting: Using data visualization to uncover initial patterns and relationship between variables.

PACE: Analyze Stage

- What steps need to be taken to perform EDA in the most effective way to achieve the project goal?

To perform EDA in the most effective way, I will follow the six steps (discover, structure, clean, join, validate, and present) with a focus on the following: Cleaning: Handling the 298 null values across ‘claim-status’ and the engagement metrics, as these are critical to ensure model prediction accuracy. Presenting: Prioritize visualization of the engagement metrics (likes, shares, and views) to check for outliers, as well as visualizing the ‘claim-status’ count against ‘verified-status’ and the ‘claim status’ count against ‘author ban status’ to determine which features will be the strongest predictors for the final model.

- Do you need to add more data using the EDA practice of joining? What type of structuring needs to be done to this dataset, such as filtering, sorting, etc.?

For thorough analysis, adding data about the author, such as the number of followers and the account history, via joining would substantially enhance the analysis further.

The type of structuring that needs to be done is data transformation, as the skilearn python library would not accept categorical variables to build predictive model.

- What initial assumptions do you have about the types of visualizations that might best be suited for the intended audience?

I presume a bar chart and histogram would be used for examining the distribution of variables; a boxplot for checking outliers; a scatterplot for identifying correlation between engagement metrics; and a pie graph to examine the proportion of a given variable.

PACE: Construct Stage

- What data visualizations, machine learning algorithms, or other data outputs will need to be built in order to complete the project goals?

The type of data visualizations that would be the most helpful for completing the project goal are a bar chart to examine the distribution of categorical data, such as claim status; a histogram to examine the distribution of continuous variables, such as engagement variables; a boxplot for checking outliers; a scatterplot for identifying correlation between engagement metrics; and a pie graph to examine the proportion of a given variable.

When it comes to the machine learning algorithms, tree-based models would be the best choice, such as random forest and XGBoost models. Other outpu will include confusion matrix and evaluation metric scores to evaluate the model performance.

- What processes need to be performed in order to build the necessary data visualizations?

To perform the necessary data visualizations, libraries like seaborn and matplotlib must be imported. Then the data must be prepared for plotting using pandas, which includes filtering, grouping, or aggregating the data to isolate the specific value being compared. The appropriate function and its parameters will be determined based on the type of graph that we need to present the findings.

- Which variables are most applicable for the visualizations in this data project?

The variables that are msot applicable for the visualization are the engagement metrics and categorical variables including “`claim_status`, `author_ban_status`, and `verified_status`.”

- Going back to the Plan stage, how do you plan to deal with the missing data (if any)?

Since the 298 missing values represent a very small fraction of the overall dataset (`19,382` rows), I plan to drop the rows containing these null values usign ‘dropna()’.

PACE: Execute Stage

- What key insights emerged from your EDA and visualizations(s)?

The key insights that emerge from the EDA and data visualization are:

Engagement metrics are heavily skewed to the right, meaning that the vast majority of the data is clustered at the lowest values, while a small fraction extends to the right. This means most videos receive low view counts, while only a small fraction receive high view counts.

The visualization shows that videos posted from verified accounts are most likely to be opinions, whereas videos posted from unverified accounts are most likely to be claims.

Active authors are more likely to post than banned and under review; however, active accounts that post claims are more likely to be banned than those that post opinions.

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- What business and/or organizational recommendations do you propose based on the visualization(s) built?

I recommended the incorporation of more data about the authors, such as the number of followers and the account history, to strengthen future model prediction. From a business operation perspective, our visualization shows that the majority of the claims, which receive 99% of the views, are posted by unverified, non-active users. I recommend an automated system that flags videos from these segments, especially if particular words were detected in the video. Additionally, by building a machine learning model and using n-gram on video transcription, we can reveal key words indicating whether a video is an opinion or claim and automate the filtering process, which improves system performance and decreases workload.

- Given what you know about the data and the visualizations you were using, what other questions could you research for the team?

Are there specific key words in the video text transcriptions that are associated with the claim videos or banned authors?

How should we handle extreme engagement outliers before we start the model training process?

- How might you share these visualizations with different audiences?

If my audience is my data team, the **visualizations** and terminology would be highly technical to facilitate idea exchange and technical suggestions. If the presentation is to the project manager, executives, or C-suite, the **visualizations** and terminology would be straightforward and easy to understand, as their main focus is on the business and operational impact. Given that, I would focus on the actionable insights more than technical processes.